CS1910 Computer Science I • Lists

Partner workbook

The Median Value

In lecture with a partner • about 20 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

The *median* of a list of numbers is the middle value once the numbers are in order. You will plan a function that finds it, starting with the questions you should ask *before* you write any code.

7	2	9	4	5	sorted →	2	4	**5**	7	9	median $= 5$

1 Understand

about 6 minutes

1. If we write this as a function, what is it called, what are its parameters, and what does it return?

2. What more do you need to know about the list before you start?

3. What should happen if the list has an **even** number of items, for example | 4 | 1 | 3 | 2 | ?

4. What should we do if the list contains strings, Booleans or other things?

2 Design and test cases

about 6 minutes

5. Complete the test cases before you write code.

Input	Expected	Normal, edge or error case?
`[3, 1, 2]`	2	normal: odd length, unsorted
`[4, 1, 3, 2]`		
`[5]`		
`[]`		
`[2, 2, 2, 9]`		
`[3, "a", 1]`		

6. Write the steps of your algorithm.

3 Implement

about 8 minutes

7. Write `median(values)`.

Before you leave

about 1 minutes

- [ ] I can decide a function's name, parameters and return type before coding.
- [ ] I can handle odd and even lengths, and the empty list.
- [ ] I know the difference between `sorted(values)` and `values.sort()`.